

Prajwal Premkumar Khairnar

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WORK EXPERIENCE

Senior Data Scientist, Jan 2025 – Present (10 months)

Data Scientist, Feb 2022 – Dec 2024 (2 years 11 months)

Northern Care Alliance NHS Foundation Trust, Manchester, U.K.

- Leading NLP and AI applications in healthcare.
- Empowered clinicians to deliver better care by developing DS/AI-ML solutions to their operational/clinical needs. Worked with internal and external stakeholders on collaborative research projects/initiatives, using Python, R and SQL.
- Working with Board Executives on broader health inequalities research to support on data-driven strategy. Experienced in senior stakeholder management and project management. Extensive experience of leading cross industry collaborative teams (public, private and research/education sectors).
- Lead AI Developer for a trust wide self-serve chat bot (~6000 users) that uses organisation specific context for answering internal questions, directing to dashboards among other offerings. Easy to use reliable source of truth & navigation for employees.
- Experienced in deep NLP research (Named Entity Recognition) using spaCy, scispaCy and LLMs (APIs and local SLMs), paired with a hybrid recommendation system for clinical coding (aiming to reduce time/patient by ~ 5 minutes). Lead author and presenter for the project at [Health and Care Analytics Conference 2023](#).
- Lead Data Scientist for Outpatient Did Not Attend Risk Predictions project. Custom internal web-app to surface results. To be presented at [Health and Care Analytics Conference 2025](#).
- Worked on clinical document summarization and classification using LLMs to enable quick and efficient information availability for clinicians.
- Cloud hosting experience (Azure, AzureML, etc.) alongside distributed computing (databricks, spark).
- Managing two data scientists, developing their skills and creating an agile environment and using git version control, to meet stakeholder needs, primarily working with Neurosurgery, Clinical Coding, A&E and Operations departments.

Founding AI Engineer, June 2024 – Present (1 year 5 months)

Rumii, London, U.K.

- Founding AI Engineer at Rumii (previously MindPsy), a mental health startup, to own development of the conversational components of the app, a compassionate co-pilot.
- End-to-end agentic LLM development from research and finetuning to validation, deployment and monitoring (LLMOps). Leading and contributing to backend architecture and system design from ground up.
- Mentoring and overseeing junior team members in a cross functional team.

NHS Clinical Entrepreneur, Jan 2024 – Present (1 year 10 months)

NHS England, U.K.

- Cohort 8 NHS Clinical Entrepreneur Fellow, selected to drive innovation in the NHS.
- The Clinical Entrepreneur Programme is an NHS initiative, delivered by Anglia Ruskin University. It is the biggest entrepreneurial workforce development programme of its kind.

Management Committee Member, Nov 2023 – Present (2 years)

NHS-R Community, U.K.

- NHS-R Community Supports the learning, application and use of R and data science tools in the health and care system in the U.K.
- The committee plays a pivotal role in shaping the approach, focus, and strategy of initiatives, fostering collective decision-making to address community needs and drive positive change.

Lead Positive Leadership Programme, July 2023 – Dec 2023 (6 months)

NHS North West Leadership Academy, Manchester, U.K.

- A bespoke leadership development programme aimed at inclusive and compassionate leadership.

INTERNSHIPS

- **EHR Development Intern (Leeds Teaching Hospitals NHS Trust)**, Aug 2021 – Feb 2022 (7 months) - Leeds, U.K.
- **Data Analyst (Intern – The Shaadi Times)**, May 2020 – July 2020 (3 months)
- **Analytics Intern (Megastar Engineering)**, Dec 2019 – June 2020 (7 months)

PROFESSIONAL SUMMARY

A proficient Senior Data Scientist/AI Engineer with around 4 years of experience, adept in AI Research, Applied AI, Natural Language Processing (NLP), Machine Learning, and advanced statistical analysis. Possesses robust proficiency in engaging with senior stakeholders and orchestrating strategic, data-informed decision-making in end-to-end project lifecycle. Demonstrated leadership acumen in fostering harmonious cross-functional collaborations and experienced in working with fast paced and early-stage start-ups as well as large organizations.

EDUCATION

MSc. Data Science and Analytics, 2021-22

University of Leeds, Leeds, United Kingdom (U.K.)
Distinction

B.E. Information Technology, 2016-20

Savitribai Phule Pune University, Pune, India
Distinction (GPA: 8.75)

PROJECT HIGHLIGHTS

- **NLP in assisting Clinical Coding (NCA NHS FT)**
NLP (NER-L) for clinical text documents. Lead author and Presenter at [Health and Care Analytics Conference 2023](#).
- **Proprietary LLM orchestration framework research**
- **LLM assisted discharge summary generation.**
- **LLM Evaluation Framework for healthcare specific use cases**
- **Sample size re-estimation in randomised clinical trials (MSc. Dissertation)**
Researched the impact of mis-specified nuisance parameters on sample size calculations in clinical trials. Demonstrated the feasibility of adjusting allocation ratios to meet financial or trial-specific needs while maintaining controlled error rates.
- **RShiny dashboard development on NHS Workforce Public Statistics**

Detailed project code and descriptions can be found on my [GitHub profile](#) and [Website](#).

CORE SKILLS

- Python, R, SQL, C, C++
- Applied AI, NLP, Deep Learning, Statistical and Multivariate Analysis, Hypothesis Testing, MLOps, LLMOps
- Pytorch, Huggingface transformers, spaCy, Data Visualization (seaborn, matplotlib, ggplot2)
- Azure (ML Studio, SQL Server, data pipelines, Databricks), Spark, PowerBI, Tableau, JIRA

REFERENCES AVAILABLE UPON REQUEST